EAB 770 - Lecture 1

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1 Lecture 1: Introduction

1.1 Outline

- Syllabus Discussion
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1.2 Syllabus

- The syllabus can be accessed through
  - Canvas
  - https://miguelfudolig-eab-770-syllabus.share.connect.posit.cloud

1.3 Programming Language

- We will primarily use R and `SAS` for this course.

  - You can access SAS through SAS On Demand for Academics. This will be free and accessible when you have internet connection.
  - You can download the most recent versions of R and RStudio if you prefer to take the course in R. You can access a quick tutorial on how to install these here.

1.4 Accessing SAS On Demand for Academics

- Create a SAS Profile that will be used to register for SAS OnDemand for Academics. To register, visit https://www.sas.com and fill out the form to *Create profile*.

  - Use your UNLV email for faster verification.

- Create your account for SAS OnDemand for Academics. To register, visit https://welcome.oda.sas.com sign in with your SAS Profile and complete the steps to *Register for an account*.

2 Why are we here?

Why bother learning categorical data analysis?

3 Categorical Data

3.1 Categorical Data

Categorical data is present everywhere!

- Demographics
- Opinions
- Ratings
- Locations

3.2 Types of Categorical Variables

3.2.1 Dichotomous/Binary

These variables can only have 0/1 values. Examples include: Heads/Tails; Alive/Dead; Selected/Not Selected

3.2.2 Nominal

These variables have more than two levels without inherent ordering. Examples include: Race; Religion; eye color.

3.2.3 Ordinal

These variables have more than two levels with inherent ordering. Examples include: Class Rank, Likert Scale

3.2.4 Discrete Counts

These variables consist of numerical outcomes consisting of counting numbers. Example include: Queue length, Number of hospitalizations

3.2.5 Grouped Survival Times

This specialized variable counts the number of patients who fail/survive during a specific time interval. The interval of failure/survival (time-to-event) is categorical.

3.3 Independent and Dependent Variables

Note

Independent variables, sometimes referred to as predictor variables, are those that are experimentally manipulated or used to predict another variable.

Note

Dependent variables, sometimes referred to as response or outcome variables, is the variable of interest as an outcome or response. **In this class, we will focus on categorical dependent variables**.

3.4 Sampling Frameworks

- Historical data
  - Observational data, usually secondary data
  - No randomization
  - Example: Public data sets.
- Experimental Data
  - Studies that involve random assignment of treatments to different subjects.
  - Random assignment/allocation
  - Example: Randomized Controlled Trials
- Sample Survey Studies
  - Randomly sampled from a larger study population
  - Random selection

3.5 Effect of Randomization

- Randomization affects conclusions of generalization and causation.
- Historical data
  - No randomization: hard to generalize unless the data was randomly sampled from a predetermined population
- Experimental Data
  - Random assignment: balances effects of confounding variables
- Sample Survey Studies
  - Random sampling: representative sample provides good coverage of the larger population

3.6 Randomization

Randomization can be done per subject or a cluster of subject.

- Strata (survey studies) or blocks (experiments)
- Examples
  - Stratified random samples
  - Randomized block designs
  - Multi-level experimental designs

3.7 `SAS` Procedures and Data

- There are multiple ways to import data into `SAS`
  - DATA step
  - PROC IMPORT step
  - `SAS` OnDemand can import data from CSV files.
- `FREQ`
  - Contingency tables (2×2, $s \times r$)
  - Chi-squared tests
- `LOGISTIC`
  - Logistic Regression
- `GENMOD`
  - Generalized Estimating Equations
- Other procedures include `GLM` and `CATMOD`.

3.8 `SAS` in Class

- Sample code will be provided, but there will be examples where we work the code out together as a class.
- Class computers have `SAS` installed in them.
  - You can save files on your Y drive (assigned to your ACE ID) to access files on other computers.
  - `SAS` can also be accessed through `SAS` OnDemand Academics.
    - Needs internet connection

3.9 `SAS` Examples

- Download Chapter1.sas from Canvas.

3.10 `SAS` Example

3.10.1 Coding Rules

- ALL lines should be ended with a semi-colon **;**
- `data` statement is followed by the data set name.
- `length` specifies the number of characters included in each **character**.

- `input` names all variables included in the data set
  - The `$` after the variable tells `SAS` that these variables are categorical variables.
- `datalines;` tells `SAS` that you are about to add new data points.
  - Each row represents a data point in this data table.
  - End with a separate `;`

3.10.2 Example

```
data respire;
length treat $7. outcome $1. ;
input treat $ outcome $ count;
datalines;
placebo f 16
placebo u 48
test f 40
test u 20
;
```

3.11 `SAS` Example: One Observation per Row

```
data severe;
input treat $ outcome $ @@;
datalines;
placebo f placebo f placebo f
placebo f placebo f
placebo u placebo u placebo u
placebo u placebo u placebo u
placebo u placebo u placebo u
placebo u
test f test f test f
test f test f test f
test f test f
test u test u test u
test u test u test u
test u test u test u
test u test u test u
test u test u test u
test u test u test u
test u test u
;
```

- This is an example of a data set where one row corresponds to one observational unit.
- The `@@` sign means that the data lines contain multiple observations. `@@` basically means new line after two entries written in the `datalines` statement.

4 R

4.1 Importing files

There are multiple ways to import files into `R`. The most common data set format is the `.csv` file. You can import CSV files using the `read.csv(filename)` function.

First, download the file `adverse.csv` from Canvas in the same folder as your R source code.

```
getwd() # save your data in this folder
adverse <- read.csv("adverse.csv") # or use a specific path to your folder
adverse
```

4.2 Data Frames

Data frames are rectangular spreadsheets. Data are typically imported as data frames.

> **Note**
>
> Rows correspond to observations and the columns correspond to variables.

Example:

```
head(mtcars)
```

```
                   mpg cyl disp  hp drat    wt  qsec vs am gear carb
Mazda RX4         21.0   6  160 110 3.90 2.620 16.46  0  1    4    4
Mazda RX4 Wag     21.0   6  160 110 3.90 2.875 17.02  0  1    4    4
Datsun 710        22.8   4  108  93 3.85 2.320 18.61  1  1    4    1
Hornet 4 Drive    21.4   6  258 110 3.08 3.215 19.44  1  0    3    1
Hornet Sportabout 18.7   8  360 175 3.15 3.440 17.02  0  0    3    2
Valiant           18.1   6  225 105 2.76 3.460 20.22  1  0    3    1
```

```
sample_data  <- data.frame(ID = c(1,2,3), Name = c("Alice","Bob","Charlie"))
sample_data
```

```
  ID    Name
1  1   Alice
2  2     Bob
3  3 Charlie
```

4.3 R packages

R packages extend the functionality of R by providing additional functions, data, and documentation. These packages are written by R users around the world and can be downloaded for free!

> **Note**
>
> Think of R as a new phone. R packages are apps that you can download to use your phone in many different ways.

4.4 Installing and Loading R Packages

Like apps on a phone, R packages also need to be installed. These packages can be installed by running the following code snippet `install.packages("PackageName")`. For example, to install the package `tidyverse` used for data manipulation and cleaning, you can run the following code:

```r
install.packages("tidyverse")
```

To load this package in R, you can use the following syntax:

```r
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

> **!** Important
>
> You must have an active internet connection to install `R` packages to your device.

5 Probability Distributions

5.1 Sampling Distributions

Recall: Inferential statistics typically aims to determine how likely it is to have obtained a sample given certain assumptions about the population the sample was drawn.

> **i** Note
>
> The sampling distribution typically depends on the properties of the underlying distribution of the random variable of interest from a population.

> **!** Important
>
> Recall: Central Limit Theorem (CLT) can be used for continuous variables. However, CLT might not work for some scenarios involving categorical response variables.

5.2 Probability Distributions for Categorical Variables

Distributions appropriate to categorical data should be used for inferential statistics.

- Hypergeometric Distribution
- Bernoulli/Binomial Distribution
- Multinomial Distribution
- Poisson Distribution

5.3 Hypergeometric Distribution

This distribution can be used for discrete variables to determine the number of "successes" in a sequence of n draws from a finite population of size M where sampling is conducted without replacement.

Hypergeometric Distribution

Suppose there are N balls in a bag and m are green. If X denotes the total number of green balls (success) desired and we have a total number of k draws from the bag, then the probability of the random variable of the total number of successes to be equal to X can be written as:

$$P(X = x) = \frac{C(m, x)C(N - m, k - x)}{C(N, k)}; x \in \{0, 1, ..., m\}$$

where $C(a, b) = \frac{a!}{b!(a-b)!}$ is the combination notation.

Note

As an example, the balls could be a pool of medical students. The green balls are international students. If we want to select a total of n students from the pool of medical students, then the probability that we will select x international students is given by the hypergeometric distribution.

5.4 Properties

The mean μ and variance σ^2 of the hypergeometric distribution can be found using the following equations:

$$\mu = \frac{km}{N}; \sigma^2 = \frac{k(m/N)(1 - m/N)(N - k)}{N - 1}$$

5.5 Hypergeometric Distribution

5.5.1 R

The probability mass function (PMF or $P(X = x)$) of the hypergeometric distribution is:

```
dhyper(x=x,
       m=m,
       n=N-m,
       k=k)
```

The cumulative distribution function (CDF or $P(X \leq x)$) of the hypergeometric distribution is:

```r
phyper(q=x,
       m=m,
       n=N-m,
       k=k)
```

5.5.2 SAS

In a data step, we can use the `pdf()` function for $(P(X=x))$, while `cdf()` can be used for $(P(X \leq x))$

```sas
data hypergeometric_prob;
p=pdf('HYPER', x, N, m, k);
c=cdf('HYPER', x, N, m, k);
;
run;
```

5.6 Example

Suppose the admissions committee at a medical college has 15 qualified applicants, 3 of which are international students, and can only admit 2 new students. What is the probability of admitting **at least one** international student if the admissions committee were to randomly select two new students from the 15 qualified applicants?

5.6.1 Distribution

We need to use the hypergeometric distribution to get P(X$\geq$ 1) where N=15, m=3, k=2. Admitting at least one international student could mean $x = 1$ or $x = 2$.

5.6.2 R

```r
N <- 15
m <- 3
k <- 2

dhyper(x=1,m=m,n=N-m,k=k) + dhyper(x=2,m=m,n=N-m,k=k)
```

```
[1] 0.3714286
```

or using complementarity, it could be 100% minus the probability of admitting no international student.

```
N <- 15
m <- 3
k <- 2

1-phyper(q=0,m=m,n=N-m,k=k)
```

```
[1] 0.3714286
```

5.6.3 SAS

```
data hypergeometric_prob;
p=pdf('HYPER', 1, 15, 3, 2)+pdf('HYPER', 2, 15, 3, 2);
c=1-cdf('HYPER', 0, 15, 3, 2);
;
run;
```

Using complementarity, it could be 100% minus the probability of admitting no international student.

5.7 Exercise

5.7.1 Question

Suppose there are one hundred classrooms in a school, twenty of which have defective audiovisual equipment. If five classrooms were randomly selected for use at a conference, what is the probability that there are NO classrooms with defective audiovisual equipment?

5.7.2 Distribution

We need to use the hypergeometric distribution to get P(X=0) where N=100, m=20, k=5.

5.7.3 R

```r
N <- 100
m <- 20
k <- 5

dhyper(x=0,m=m,n=N-m,k=k)
```

```
[1] 0.3193094
```

5.7.4 SAS

```sas
data hypergeometric_prob;
p=pdf('HYPER', 0,100,20,5);
;
run;
```

5.8 Bernoulli Distribution

This distribution can be used for binary variables, i.e. variables that can only take on two values (0/1). This distribution can be used to determine the probability of success if only one draw is made from a finite population. If there are m successes in a total of N possible outcomes, the probability of drawing a "success" in a single trial can be written as:

$$P(X = 1) = m/N$$

It follows that the probability of failure is $P(X = 0) = 1 - m/N$.

5.9 Binomial Distribution

This distribution can also be used for binary variables, however, the binomial distribution deals with more than one trial (denoted by n).

> **Note**
>
> The binomial distribution assumes that the probability of success does not change after every trial, i.e. sampling with replacement. Each trial is treated as independent of the others, unlike the hypergeometric distribution where sampling is conducted without replacement, effectively changing the probability of success after every draw.

The probability of observing x successes for a series of n trials with a probability of success p according to the binomial distribution can be written as:

$$P(X = 1) = C(n, x)p^x(1 - p)^{(n-x)}; x \in \{0, 1, 2, ..., n\}$$

5.10 Properties

The mean and variance of the binomial distribution can be expressed as:

$$\mu = np; \sigma^2 = np(1 - p)$$

5.11 Binomial Distribution

5.11.1 R

The PMF can be obtained using the `dbinom` function, while the CDF can be obtained using the `pbinom` function.

```
dbinom(x=x,size=n,prob=p)
pbinom(q=x,size=n,prob=p,lower.tail=TRUE)
```

> **Note**
>
> `lower.tail=FALSE` tells `R` that we are interested in $P(X > x)$ instead of $P(X \leq x)$

5.11.2 SAS

In a data step, we can use the `pdf()` function for $(P(X = x))$, while `cdf()` can be used for $(P(X \leq x))$

```
data binomial_prob;
p=pdf('BINOMIAL', x,p, n);
c=cdf('BINOMIAL', x, p, n);
;
run;
```

5.12 Example

Suppose that the probability of being proficient in mathematics at a particular school is 0.7. Three students are chosen at random from this school. What is the probability that all three are proficient in mathematics?

5.12.1 Distribution

We need to use a binomial distribution with a probability of success $p = 0.7$ out of a total sample size $n = 3$. We need to solve for $P(X = 3)$.

5.12.2 `R`

```
p<- 0.7
n<- 3

dbinom(x=3,size=n,prob=p)
```

```
[1] 0.343
```

5.12.3 `SAS`

```
data binomial_prob;
p=pdf('BINOMIAL', 3,0.7,3);
;
run;
```

5.13 Exercise

Suppose that the survival probability of a new surgical procedure was 0.85. If a total of 200 patients went through the procedure independently, what is the probability that more than 190 of these patients survived the procedure?

5.13.1 Distribution

We can consider the random variable to be those who survived the procedure. need to use a binomial distribution with a probability of success $p = 0.85$ out of a total sample size $n = 200$. We need to solve for $P(X > 190)$.

5.13.2 R

```
p<- 0.85
n<- 200

1-pbinom(q=190,size=n,prob=p)
```

```
[1] 2.005539e-06
```

```
#OR
pbinom(q=190,size=n,prob=p,lower.tail=FALSE)
```

```
[1] 2.005539e-06
```

5.13.3 SAS

```
data binomial_prob;
p=1-cdf('BINOMIAL', 190,0.85,200);
;
run;
```

5.14 Multinomial Distribution

This distribution can be used to model random variables wih more than two possible outcomes. In a multinomial distribution, each trial results in one of I outcomes, where I is some fixed finite number and the probability of each possible outcome can be expressed by $p_1, p_2, ...p_I$ such that the sum of all probabilities is $\sum_{i=1}^{i} p_i = 1$.

> **Example**
>
> If students were classified as proficient or not proficient, the binomial distribution would be appropriate to use. However, if students were classified with more levels like basic, intermediate, and advanced proficiency, the multinomial distribution would need to be used.

5.15 Multinomial Distribution: PMF

For n trials with I possible outcomes, where p_i is the probability of the i^{th} outcome, the joint probability that each possible outcome will have value x_i is:

$$P(X_1 = x_1, X_2 = x_2, ..., X_I = x_I) = \frac{n!}{x_1!x_2!...x_I!} p_1^{x_1} p_2^{x_2} ... p_I^{x_I}$$

with the constraint that $\sum_i x_i = n$.

5.16 Properties

For the i^{th} outcome, the mean can be expressed similar to the binomial distribution.

$$\mu_i = np_i; \sigma^2 = np_i(1 - p_i)$$

5.17 Multinomial Distribution

The following functions can be used to determine the value of the PMF of the multinomial distribution

5.17.1 `R`

```
x <- c(x1,x2,...,xI)
p <- c(p1,p2,...,pI)
dmultinom(x=x,size=n,prob=p)
```

5.17.2 `SAS`

There is no standalone `PDF` function in SAS.

```
data multinom_prob;
p=fact(n)/(fact(x1)*fact(x2)*fact(x3)*[...])*p1**x1*p2**x2*p3**x3*[...];
;
run;
```

5.18 Example

Suppose exposure levels to secondhand smoke for a particular community was classified into three levels: "none","light", and "heavy". If the probability of having no exposure (none) is 0.05 and having light exposure is 0.37, what is the probability that out of 10 community members, 2 had no exposure, 5 had light exposure, and the rest had heavy exposure?

5.18.1 Distribution

Because there are three possible outcomes for exposure levels, we need to use the multinomial distribution. For this problem, $n = 10$, $x_1 = 2, x_2 = 5, x_3 = 3$. The sum of the probabilities should be 1, so the probabilities are $p_1 = 0.05, p_2 = 0.37, p_3 = 0.58$.

5.18.2 `R`

```
n<- 10
x <- c(2,5,3)
p <- c(0.05,0.37,0.58)
sum(p)
```

```
[1] 1
```

```
dmultinom(x=x,size=n,prob=p)
```

```
[1] 0.008523798
```

5.18.3 `SAS`

```
data multinom_prob;
p=fact(10)/(fact(3)*fact(5)*fact(2))*0.05**2*0.37**5*0.58**3;
;
run;
```

5.19 Example

Suppose that for a particular school district, the probability of having being a minimal reader is 0.12, the probability of being a basic reader is 0.23, and the probability of being a proficient reader is 0.65. If six students were selected at random, there will be two students from each category.

5.19.1 Distribution

Because there are three possible outcomes for reading level, we need to use the multinomial distribution. For this problem, $n = 6$, $x_1 = 2, x_2 = 2, x_3 = 2$. The sum of the probabilities should be 1, which is satisfied by the probabilities. $p_1 = 0.12, p_2 = 0.23, p_3 = 0.65$.

5.19.2 R

```
n<- 6
x <- c(2,2,2)
p <- c(0.12,0.23,0.65)
sum(p)
```

```
[1] 1
```

```
dmultinom(x=x,size=n,prob=p)
```

```
[1] 0.02896592
```

5.19.3 SAS

```
data multinom_prob;
p=fact(n)/(fact(x1)*fact(x2)*fact(x3)*[...])*p1**x1*p2**x2*p3**x3*[...];
;
run;
```

5.20 Poisson Distribution

This distribution is often used to model count data that varies randomly over time.

> **Note**
>
> As the number of trials gets larger, the binomial distribution tends to converge to the Poisson distribution when p is fixed. The binomial distribution assumes that the number of trials is fixed, while the Poisson distribution assumes the period of time in which the observations occur must be fixed.

If the expected/average number of successes expected to occur in a fixed interval of time is λ, the probability of observing x successes in that time interval can be written using the Poisson PMF:

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}; x \in \{0, 1, 2, ...\}$$

5.21 Properties

The mean and variance of the Poisson distribution are equal.

$$\mu = \lambda; \sigma^2 = \lambda$$

Tip

With real-world count data, the variance often exceeds the mean. We refer to this phenomenon as overdispersion.

Note

For sufficiently large λ, the normal distribution can approximate the Poisson distribution.

5.22 Poisson Distribution

5.22.1 R

The functions `dpois()` and `ppois()` can be used to generate the PMF and CDF of the Poisson distribution, respectively.

```
dpois(x=x,lambda=lambda)
ppois(q=x,lambda = lambda,lower.tail=TRUE/FALSE)
```

5.22.2 SAS

```
data pois_prob;
p=pdf('POISSON',x,lambda);
c=cdf('POISSON',x,lambda);
;
run;
```

5.23 Example

Suppose that over the past month, the hourly average number of patients who use the restroom at the waiting room of a particular clinic is 4.2. What is the probability that there will be 2-3 patients using the restroom in the next hour?

5.23.1 Distribution

We can use the Poisson distribution to model the hourly count of patients using the restroom. The hourly average is 4.2, which serves as λ.

5.23.2 `R`

```
lambda <- 4.2
dpois(x=2,lambda=lambda) + dpois(x=3,lambda=lambda)
```

```
[1] 0.3174264
```

5.23.3 `SAS`

```
data pois_prob;
p=pdf('POISSON',2,4.2) + pdf('POISSON',3,4.2) ;
;
run;
```

5.24 Exercise

5.24.1 Question

Suppose that over the last 50 years the average number of suicide attempts that occurred at a particular correctional facility each year is approximately 2.5. What is the probability that there are:

- Exactly five suicide attempts at the facility in the next year.
- At least ten suicide attempts at the facility in the next year.

5.24.2 R

```
# Exactly 5
lambda<- 2.5
dpois(5,lambda=lambda)
```

```
[1] 0.06680094
```

```
# At least 10 = 10 and higher

1-ppois(q=9,lambda=lambda)
```

```
[1] 0.0002773521
```

```
ppois(q=9,lambda=lambda,lower.tail = FALSE)
```

```
[1] 0.0002773521
```

5.24.3 SAS

```
data pois_prob;
p1=pdf('POISSON',5,2.5)  ;
p2=1-cdf('POISSON',9,2.5)
;
run;
```